HPC ACCELERATED AMR PHENOTYPE PREDICTION

23AID205 Introduction to AIML | 23BI0112 Introduction to Biological Data
23AID304 High Performance Cloud Computing

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INTRODUCTION

- Antimicrobial Resistance (AMR) is a major public-health challenge.
- Bacterial genomes contain information associated with resistance
- Machine Learning can learn patterns from genome sequences to predict experimentally observed resistance.
- Genome-scale processing becomes computationally expensive as the number of bacterial isolates increases.
- Efficient processing is required to handle large-scale genomic data.

REVIEW CONTENTS

A genome-sequence research prototype for gentamicin phenotype labels

231	98 / 133	78.7%	18 / 18
eligible genomes	Resistant / Susceptible	best held-out accuracy	HPC parity checks

01 Cohort and sequence QC

02 K-mer and descriptor features

03 Serial / OpenMP / MPI benchmarking

04 Training CV and held-out results

05 AMRFinderPlus findings and prototype scope

Data source

Question

Can genome-derived features distinguish the recorded gentamicin Resistant and Susceptible labels in eligible P. aeruginosa isolates?

ORGANISM

Pseudomonas aeruginosa (taxon 287)

SOURCE

BV-BRC genome_amr public API
https://www.bv-brc.org/api
https://www.bv-brc.org/api/genome_amr/
https://www.bv-brc.org/api/genome_sequence/

OUTCOME

Resistant or Susceptible

ANTIBIOTIC

Gentamicin

EVIDENCE FILTER

Laboratory Method

AVAILABLE COHORT

231 eligible genomes

Source: cohort/reconciliation_summary.json; cohort/cohort_manifest.csv

Cohort curation and class balance

288
BV-BRC gentamicin source records

→

276
unique genome IDs

→

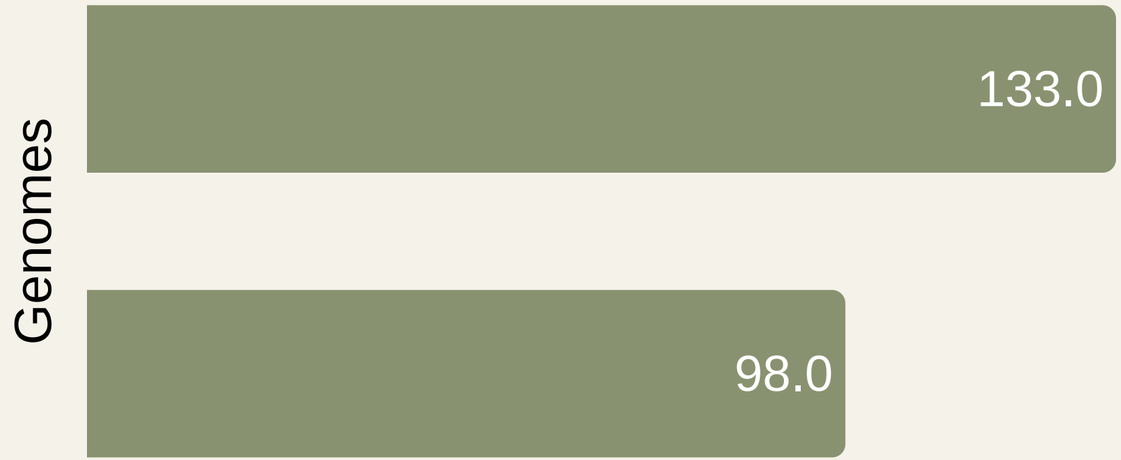
231
eligible Resistant / Susceptible genomes

After deduplication and eligibility checks

Removed from the unique records

30 missing or conflicting phenotype records

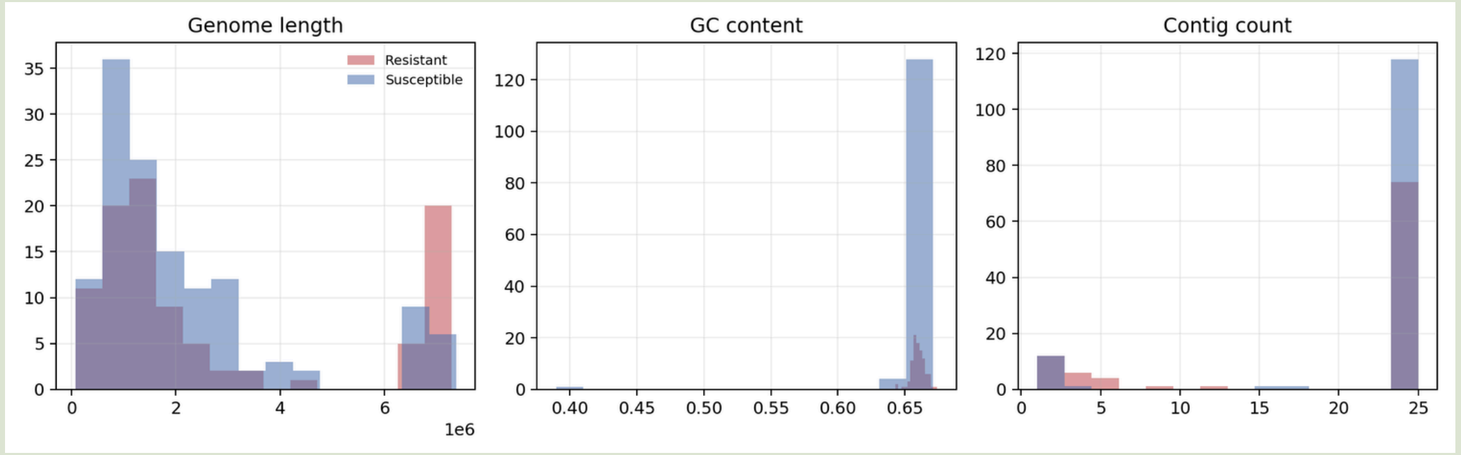
15 non-binary phenotype records



Resistant 42.4% · Susceptible 57.6%

Source: cohort/reconciliation_summary.json

Sequence QC and genome composition



231 / 231

matching FASTAs passed sequence checks

0

empty FASTA files

QC records preserve genome ID, sequence length, contig count, alphabet validity and SHA-256.

Source: qc/download_qc_status.csv; features/sequence_descriptors.csv

Feature generation

Canonical reverse-complement k-mers

Each genome becomes a numeric vector of k-mer counts or frequencies. Reverse-complement pairs share one feature.

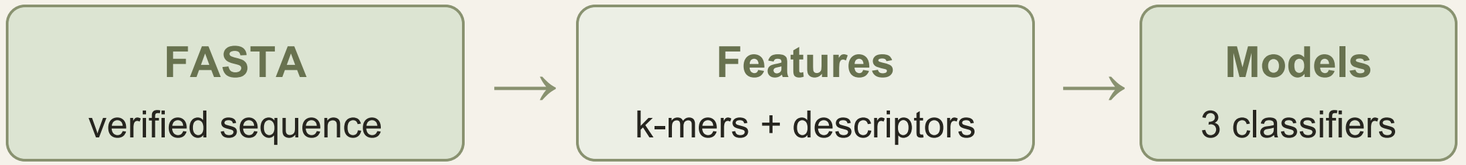
k	Features
3	32
4	136
5	512
6	2,080
7	8,192
8	32,896

Sequence descriptors · 111

Length · GC and AT skew · ambiguity · Shannon entropy
A/C/G/T composition · di- and trinucleotide frequencies
Maximum homopolymer · sliding-window GC variation

AMRFinderPlus determinant indicators · 214

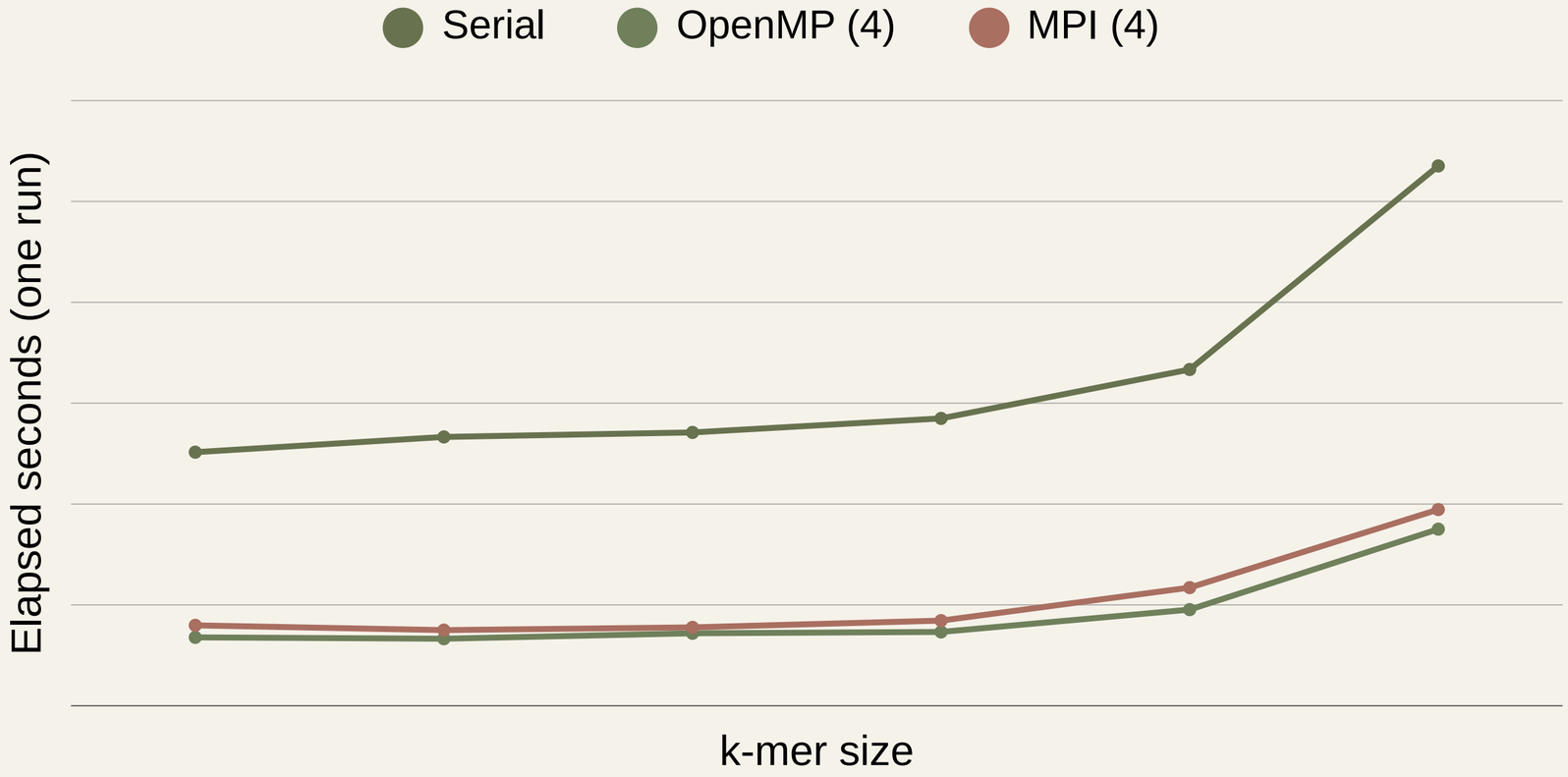
Included as a candidate feature family in the later exploratory comparison. It is not a phenotype label. The original local k=6 inference models remain sequence-only.



Source: features/sequence_descriptors.csv; feature_schema.json; amrfinder_feature_presence.csv

Serial, OpenMP and MPI feature extraction

Same 231 FASTAs · k = 3–8 · 4 OpenMP workers / 4 MPI ranks



At k = 6

Serial 14.24 s

OpenMP 3.66 s

MPI 4.22 s

3.89×

OpenMP / Serial at k=6

18 / 18

implementation-k outputs matched Serial exactly (max difference 0)

Measured on one WSL2 host; one timing per implementation and k. HPC accelerated feature extraction, not model fitting.

Source: hpc/timings_by_k.csv; hpc/correctness_by_k.csv; run_manifest.json

MODEL EVALUATION

Training and evaluation design

231 genomes
98 Resistant
133 Susceptible

→

Train · 184
Stratified 5-fold CV
shared folds for comparisons

+

Test · 47
Locked holdout
used after CV selection

Models

Logistic Regression · Random Forest · XGBoost

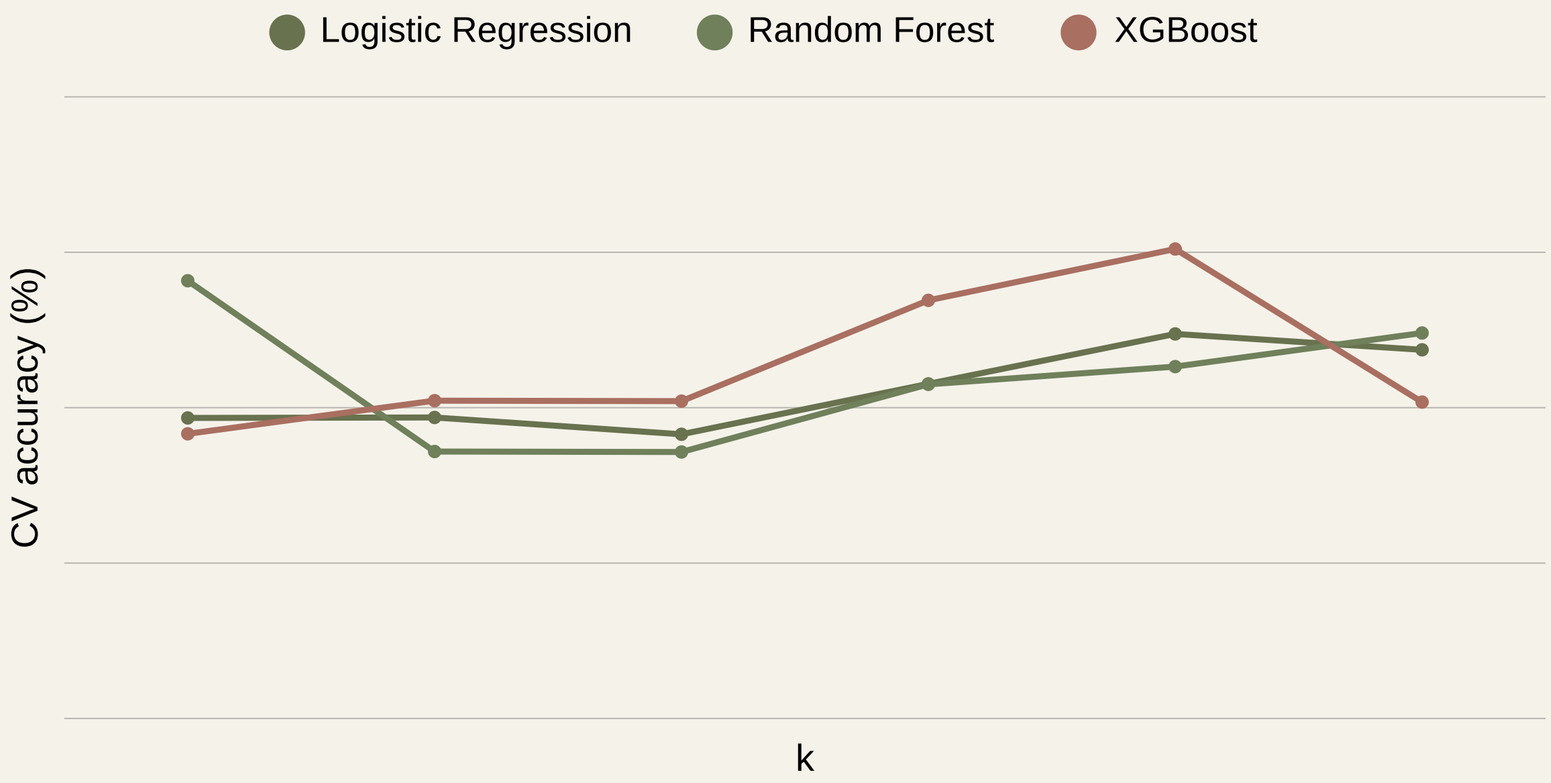
27 candidate feature sets × 3 models = 81 training-CV configurations

- We chose each model's feature set using only the 184 training genomes, based on its average Resistant F1 across five CV folds.
- We kept the 47 test genomes aside until the feature choices were finished, so they did not influence selection.
- HPC sped up making the k-mer features. The model training itself ran on the computer's CPU using scikit-learn and XGBoost.

Source: split_manifest.csv; cv_fold_metrics.csv; run_manifest.json

K-mer-size sensitivity

Canonical k-mer features only · mean training-CV accuracy across five folds



Highest accuracy among k-only sets

Model	k	Accuracy ± fold SD
Logistic Regression	7	67.4% ± 5.2%
Random Forest	3	69.1% ± 6.3%
XGBoost	7	70.1% ± 3.0%

The strongest k differs by model. k=8 is not a universal winner; accuracy and Resistant F1 do not peak at the same k for every classifier.

Each point is the mean of five training folds. The table reports fold standard deviation at the accuracy-leading k.

Source: cv_feature_set_summary.csv; canonical_k_cv_summary.csv

Feature-set comparison on training CV

Each model selected its feature set by mean Resistant F1 across training folds.

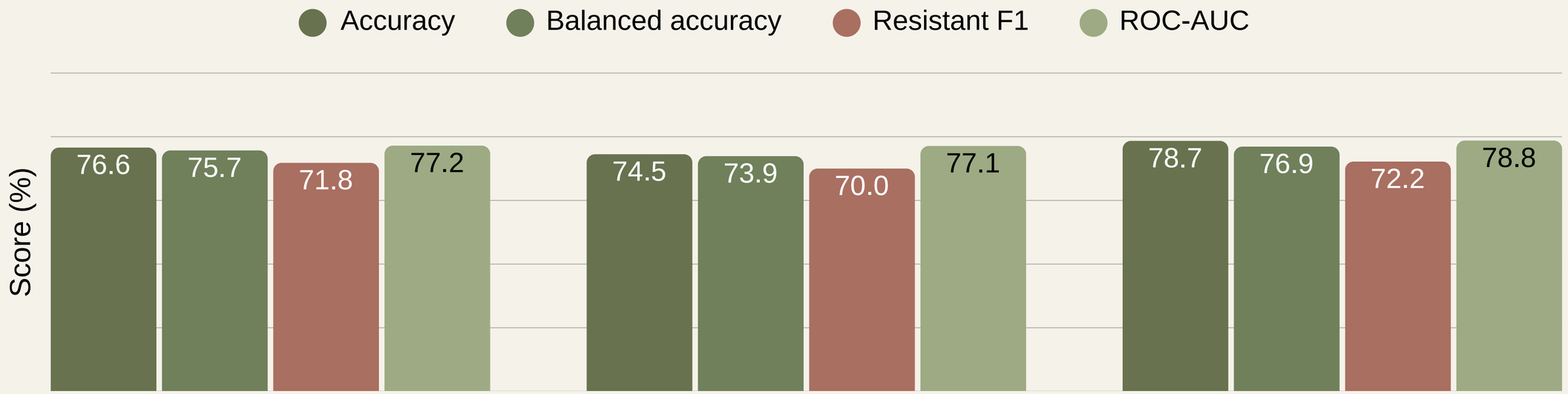
Model	Selected candidate features	CV accuracy mean ± SD
Logistic Regression	canonical k6 plus amrfinder	74.5% ± 2.6%
Random Forest	amrfinder determinants	73.9% ± 4.9%
XGBoost	amrfinder determinants	74.0% ± 6.4%

What was compared

Canonical k-mer sets (k=3–8), 111 sequence descriptors, 214 AMRFinderPlus determinant indicators, and combinations of those groups.

Source: cv_feature_set_summary.csv; heldout_metrics.csv

Held-out test results · 47 genomes



Logistic Regression
Predicted R 19 | S 28

	Predicted S	Predicted R
Actual S	22	5
Actual R	6	14

Random Forest
Predicted R 20 | S 27

	Predicted S	Predicted R
Actual S	21	6
Actual R	6	14

XGBoost
Predicted R 16 | S 31

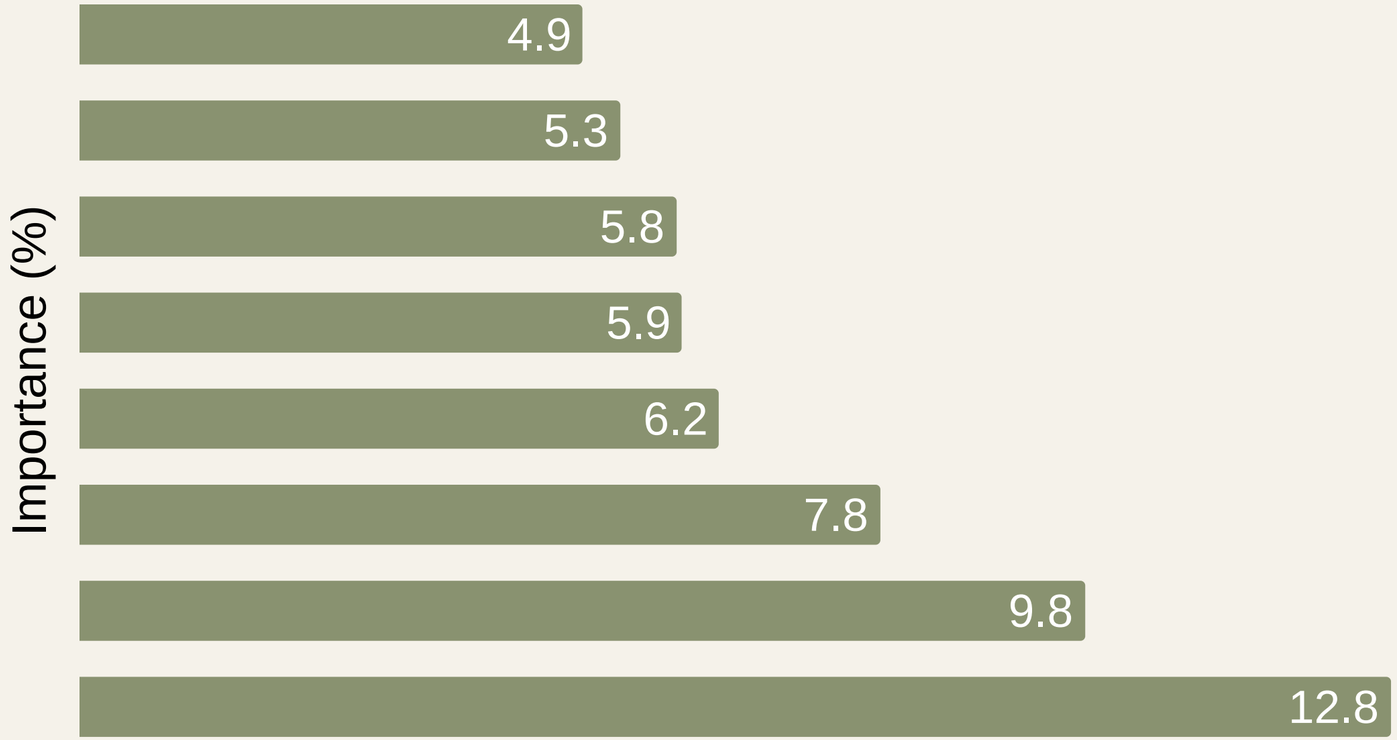
	Predicted S	Predicted R
Actual S	24	3
Actual R	7	13

Source: heldout_metrics.csv; heldout_predictions.csv

BIOLOGICAL INTERPRETATION

What the selected XGBoost model used

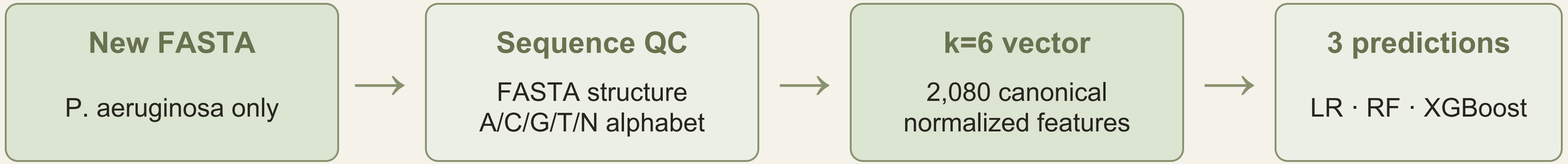
The expanded comparison selected the 214-feature AMRFinderPlus determinant set for XGBoost by training-CV Resistant F1.



Top signals include sul1, merD, merA and ant(2″)-Ia. These span multiple antimicrobial / stress classes, not only gentamicin.

Source: top_feature_importance.csv; selected feature family from heldout_metrics.csv

Current inference workflow



What a new input receives

Local command-line scoring for an unseen P. aeruginosa genome, with predicted label and Resistant probability from saved k=6 models.

Source: scripts/predict_lab896_fasta.py; web/ static project outputs

Conclusions

01 A real, curated cohort

231 eligible BV-BRC laboratory-method records were retained from 288 source records; 98 Resistant and 133 Susceptible.

02 HPC reduced feature-extraction time

At k=6, OpenMP took 3.66 s versus 14.24 s serial (3.89× on this run); all 18 output checks matched exactly.

03 Performance is promising

XGBoost led this 47-genome holdout at 78.7% accuracy and 0.788 ROC-AUC; the sample is too small for broad or clinical claims.

Source: fresh cohort manifest; heldout_metrics.csv; hpc/correctness_by_k.csv

Reference

01 BV-BRC genome_amr API · gentamicin · taxon 287 · Laboratory Method

02 AMRFinderPlus database 2026-08-07.1 · per-genome reports and consolidated hit table

03 Kokot M, Długosz M, Deorowicz S (2017). KMC 3: counting and manipulating k-mer statistics. Bioinformatics 33(17):2759–2761. doi:10.1093/bioinformatics/btx304

04 Mahé P, Tournoud M (2018). Predicting bacterial resistance from whole-genome sequences using k-mers and stability selection. BMC Bioinformatics 19:383. doi:10.1186/s12859-018-2403-z

05 Olson RD et al. (2023). Introducing the Bacterial and Viral Bioinformatics Resource Center (BV-BRC). Nucleic Acids Research 51(D1):D678–D689. doi:10.1093/nar/gkac1003

Core outputs

cohort/cohort_manifest.csv · ml/heldout_metrics.csv · cv_feature_set_summary.csv · hpc/timings_by_k.csv · hpc/correctness_by_k.csv · eda/

Source: project manifests and published sources